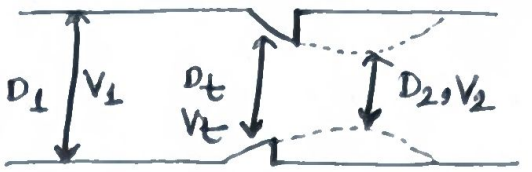
• ASSUMPTIONS: STEADY, INCOMPRESSIBLE, ALONG STREAMLINE, FRICTIONLESS, $z_1 = z_2$

• CONSERVATION OF MASS ⇒ $V_1 D^2 = V_t D_t^2 = V_2 D_2^2$

• BERNOULLI'S EQN: $p_1/\rho + \alpha_1 V_1^2/2 = p_2/\rho + \alpha_2 V_2^2/2 \Rightarrow V_2 = \sqrt{\dfrac{2(p_1-p_2)}{\rho\{1-(A_2/A_1)^2\}}}$



$\Rightarrow \dot{m}_{THEORETICAL} = \rho V_2 A_2 = \dfrac{A_2}{\sqrt{1-(A_2/A_1)^2}} \times \sqrt{2\rho(p_1-p_2)} \propto \sqrt{\Delta p}$

• SINCE A_2 IS UNKNOWN, CALCULATIONS ARE TYPICALLY DONE WITH A_t, WITH A DISCHARGE COEFFICIENT

$\dot{m}_{ACTUAL} = \dfrac{C A_t}{\sqrt{1-(A_t/A_1)^2}} \sqrt{2\rho(p_1-p_2)}$; C = DISCHARGE COEFFICIENT

• ALTERNATIVE FORM, $\beta = D_t/D_1 \Rightarrow A_t/A_1 = \beta^2 \Rightarrow \dot{m}_{ACTUAL} = C A_t \sqrt{\dfrac{2\rho(p_1-p_2)}{1-\beta^4}} = K A_t \sqrt{2\rho(p_1-p_2)}$ K = FLOW COEFFICIENT.

• ORIFICE METER: (A) COEFFICIENT OF VELOCITY = (ACTUAL VELOCITY OF JET)/(THEORETICAL VELOCITY) = C_v

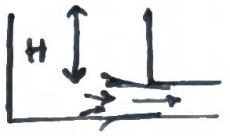
(B) COEFFICIENT OF CONTRACTION = (AREA OF JET AT VENA CONTRACTA)/(ORIFICE AREA) = C_c

(C) COEFFICIENT OF DISCHARGE = (ACTUAL FLOW RATE)/(THEORETICAL FLOW RATE) = $C_d = C_v \times C_c$

• MOUTHPIECE RUNNING FREE: (HAPPENS FOR SHORT TUBES)

$p = \rho g H \Rightarrow F = \rho g h \cdot a = \rho a_c V_c^2 \Rightarrow H = \dfrac{a_c}{a} \cdot \dfrac{V_c^2}{g}$ (MOMENTUM)

BERNOULLI, $H = V_c^2/2g \Rightarrow a_c/a = 1/2$, $V_c/V_a = 1 \Rightarrow C_d = V_2$



• MOUTHPIECE RUNNING FULL: (HAPPENS FOR LONGER TUBES)

$H = V^2/2g + h_e$; $h_e = \dfrac{(V_c - V)^2}{2g} \Rightarrow H = \dfrac{V^2}{2g} + \dfrac{V_c^2 + V^2 - 2VV_c}{2g}$

~~$a_c/a = 1 \Rightarrow H = V_c^2/g \Rightarrow V^2/2g + V_c^2/2g - VV_c/g \Rightarrow V_c^2/2g = V^2/2g - VV_c/g$~~

CONTINUITY ⇒ $a_c V_c = aV \Rightarrow V_c = V \Rightarrow V = \sqrt{gH} \Rightarrow C_v = \sqrt{gH}/\sqrt{2gH} = 1/\sqrt{2}$

EXAMPLE: A LARGE TANK HAS A SHARP-EDGED CIRCULAR ORIFICE WITH A DIAMETER OF 30 mm, LOCATED 4m BELOW A CONSTANT WATER LEVEL. THE JET ISSUES HORIZONTALLY, AND AFTER 2.65m IT FALLS BY 0.5m. THE MEASURED DISCHARGE IS 4 l/s. DETERMINE THE COEFFICIENTS OF VELOCITY, DISCHARGE & CONTRACTION.

$A = \pi/4 \times D^2$, $V_{TH} = \sqrt{2gH} \Rightarrow Q_{TH} = A \cdot V_{TH} \approx 0.006262$, $Q_{ACT} = 0.004 \Rightarrow C_d = 0.639$,

PROJECTILE MOTION ⇒ $x = Vt$, $y = \frac{1}{2}gt^2 \Rightarrow V = \sqrt{gx^2/2y} \approx 8.3 m/s \Rightarrow C_v = 8.3/\sqrt{2gH} \approx 0.937$, $C_d = C_v \cdot C_c \Rightarrow C_c \approx 0.682$

AS WE PROGRESS ALONG THE LENGTH, TURBULENT BL BECOMES MORE PROMINENT ⇒ $Re_x = \rho V x/\mu$

• SHEAR STRESS PROFILES OF TURBULENT BL'S ARE SIGNIFICANTLY DIFFERENT.

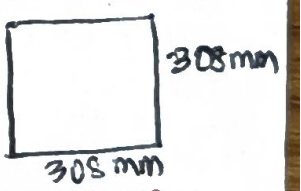
• $\dot{m}_{ACTUAL} = \int \rho u \cdot w dy$, $\dot{m}_{WITHOUT\ BL} = \int_0^\infty \rho U w dy \Rightarrow \dot{m}_{LOST} = \int_0^\infty \rho (U-u) w dy = \rho U \delta^* w \Rightarrow \delta^* = \int_0^\infty (1-u/U) dy$

(TYPICALLY ONLY UPTO δ AS VELOCITY BECOMES SAME)

• $P_{ACTUAL}^{FLUX} = \int_0^\infty \rho u^2 w dy$, $P_{NO\ BL}^{FLUX} = \int_0^\infty \rho u \cdot U w dy \Rightarrow P_{LOST}^{FLUX} = \int_0^\infty \rho u (U-u) w dy = \rho U^2 \theta w \Rightarrow \theta = \int_0^\infty u/U\{1-u/U\} dy$

(AGAIN, TYPICALLY ONLY UP TO δ^* AS INTEGRAL VANISHES)

EXAMPLE: A LAB WIND TUNNEL HAS A TEST SECTION THAT IS 308mm SQUARE. BL-VELOCITY PROFILES ARE MEASURED AT TWO CROSS SECTIONS AND DISPLACEMENT THICKNESSES ARE EVALUATED FROM THE PROFILES. AT SECTION ①, $U_1 = 28 m/s$, $\delta_1^* = 1.8$ mm. AT SECTION ② DOWNSTREAM, $\delta_2^* = 2.4$mm. CALCULATE CHANGE IN STATIC PRESSURE B/W ① & ②.



@ SECTION 1, EQUIVALENT SIDE = $(308 - 2\times1.8) \Rightarrow A_1 = (308-2\times1.8)^2 \times 10^{-6}$

@ SECTION 2, EQUIVALENT SIDE = $(308 - 2\times2.4) \Rightarrow A_2 = (308-2\times2.4)^2 \times 10^{-6}$ } IN THIS, WE HAVE UNIFORM FLOW.

BERNOULLI ⇒ $p_1/\rho + V_1^2/2 = p_2/\rho + V_2^2/2$, CONTINUITY ⇒ $A_1V_1 = A_2V_2 \Rightarrow V_2 = A_1/A_2 \times V_1$

$\Rightarrow \left(\dfrac{p_1-p_2}{\frac{1}{2}\rho V_1^2}\right) = \left\{\dfrac{308-2\times1.8}{308-2\times2.4}\right\}^4 - 1 \approx 1.6\%$

• $u \to U$ as $y \to \delta$, $\partial u/\partial y = 0$ AS $y = \delta$, SCALING

• COM $\partial u/\partial x + \partial v/\partial y = 0$, COLM SCALING, $U/L \sim V/\delta \Rightarrow V = U\delta/L$ } SCALING FOR Y-VELOCITY. ZERO-PRESSURE GRADIENT ⇒ FLAT

• COLM-x: $u\partial u/\partial x + v\partial u/\partial y = \nu\{\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2\}$, COLM-y: $u\partial v/\partial x + v\partial v/\partial y = -\frac{1}{\rho}\partial p/\partial y + \nu\{\partial^2 v/\partial x^2 + \partial^2 v/\partial y^2\}$

$U^2/L\{u\partial u/\partial x + v\partial u/\partial y\} = \dfrac{\nu U}{\delta^2}\left\{\dfrac{\delta^2}{L^2}\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2}\right\}$, $\dfrac{U^2\delta}{L^2}\left\{u\dfrac{\partial v}{\partial x} + v\dfrac{\partial v}{\partial y}\right\} = -\dfrac{U^2}{\delta}\dfrac{\partial p}{\partial y} + \dfrac{\nu U\delta}{L\delta^2}\left\{\dfrac{\delta^2}{L^2}\dfrac{\partial^2 v}{\partial x^2} + \dfrac{\partial^2 v}{\partial y^2}\right\}$

$\Rightarrow u\dfrac{\partial u}{\partial x} + v\dfrac{\partial u}{\partial y} = \dfrac{\nu L}{U\delta^2} \times \dfrac{\partial^2 u}{\partial y^2}$, $\dfrac{\delta}{L}\left\{u\dfrac{\partial v}{\partial x} + v\dfrac{\partial v}{\partial y}\right\} = -\dfrac{L}{\delta}\cdot\dfrac{\partial p}{\partial y} + \dfrac{1}{Re}\cdot\dfrac{L}{\delta} \times \dfrac{\partial^2 v}{\partial y^2} \Rightarrow \dfrac{\partial p}{\partial y} = 0$

• BLASIUS: $u/U = g(\eta)$, WHERE $\eta = y/\delta$ (AND) $\delta \propto \sqrt{\nu x/U}$, THEREBY SETTING $\eta = y\sqrt{U/\nu x}$.

• STREAMFUNCTION: $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$ { NON-DIMENSIONALISE USING $[U][\delta] = [\psi] \Rightarrow \Psi = \dfrac{\psi\sqrt{U}}{U\sqrt{\nu x}} = \dfrac{\psi}{\sqrt{U x \nu}} = f(\eta)$

• COM-SATISFIED AUTOMATICALLY, COLM: $u\partial u/\partial x + v\partial u/\partial y = \nu\partial^2 u/\partial y^2$

- $u = \frac{\partial \psi}{\partial y} = \frac{\partial \psi}{\partial \eta}\cdot\frac{\partial \eta}{\partial y} = U\frac{df}{d\eta}$, $v = -\frac{\partial \psi}{\partial x} = -\left\{\sqrt{\nu x U}\cdot\frac{\partial f}{\partial \eta}\cdot\frac{\partial \eta}{\partial x} + f\cdot\frac{\sqrt{\nu U}}{2\sqrt{x}}\right\} = \frac{1}{2}\sqrt{\frac{\nu U}{x}}\left\{\eta\frac{df}{d\eta} - f\right\}$
- $\frac{\partial u}{\partial x} = -\frac{U}{2x}\eta\frac{d^2f}{d\eta^2}$, $\frac{\partial u}{\partial y} = U\sqrt{U/\nu x}\frac{d^2f}{d\eta^2}$, $\frac{\partial^2 u}{\partial y^2} = \frac{U^2}{\nu x}\frac{d^3f}{d\eta^3} \Rightarrow 2\frac{d^3f}{d\eta^3} + f\frac{d^2f}{d\eta^2} = 0$
- AT $\eta = 5$, $u/U \approx 0.9912$ $\Rightarrow$ y CAN BE USED FOR BL THICKNESS $\Rightarrow$ $\delta = 5x/\sqrt{Re_x}$
- WALL SHEAR STRESS $= \mu\frac{\partial u}{\partial y}\Big|_{y=0} = \frac{d^2f}{d\eta^2}\mu U\sqrt{\frac{U}{\nu x}}\Big|_{y=0} = 0.332\,\mu U\sqrt{\frac{U}{\mu x}\cdot\rho} = \frac{0.332\,\rho U^2}{\sqrt{Re_x}}$; $C_f = \frac{\tau_w}{\frac{1}{2}\rho U^2} = \frac{0.664}{Re_x}$

- STEADY ASSUMPTION IS VALID BECAUSE THE EQUATIONS (MOMENTUM INTEGRAL) ARE WRITTEN FOR TIME-AVERAGED COMPONENTS OF VELOCITY. (HENCE ALSO USED TO ANALYSE TURBULENT PROFILES).

- COM: $\cancel{\frac{\partial}{\partial t}\int_{CV}\rho\,d\forall}^{=0\ (STEADY)} + \int_{CS}\rho\vec{V}\cdot d\vec{A} = 0 \Rightarrow \dot{m}_{ab} + \dot{m}_{bc} + \dot{m}_{cd} = 0$; $\dot{m}_{ab} = \int_0^\delta -\rho u w\,dy$, $\dot{m}_{cd} = \int_0^\delta \rho u w\,dy + \left\{\frac{\partial}{\partial x}\int_0^\delta \rho u w\,dy\right\}dx$
  $\Rightarrow \dot{m}_{bc} = -w\left\{\frac{\partial}{\partial x}\int_0^\delta \rho u\,dy\right\}\cdot dx$
- COLM-x: $F_{SURF,x} + \cancel{F_{BODY,x}}^{=0\ (GRAVITY\ NOT\ ALONG\ x)} = \frac{\partial}{\partial t}\int_{CV}\rho u\,d\forall^{=0\ (STEADY)} + \int_{CS}\rho u(u\cdot dA)$
- WE KNOW FROM DIFFERENTIAL ANALYSIS THAT PRESSURE ONLY VARIES ALONG x. $\begin{cases} F_{ab} = w\cdot\delta\,\bar{p} \\ F_{cd} = \left\{p + \frac{\partial p}{\partial x}\cdot dx\right\}w\left\{\delta + d\delta\right\} \\ F_{bc} = \left\{p + \frac{\partial p}{\partial x}\cdot\frac{dx}{2}\right\}\cdot w\cdot d\delta \end{cases}$ (AVERAGE PRESSURE)
- MOMENTUM FLUX: $ab = -\int_0^\delta \rho u^2 dy\,w$, $bc = -Uw\,dx\left\{\frac{\partial}{\partial x}\int_0^\delta \rho u\,dy\right\}$, (FREE STREAM VELOCITY, NOT VARYING IN y).

$$cd = \int_0^{\delta+d\delta}\rho u^2 dy\cdot w + \frac{\partial}{\partial x}\left\{\int_0^{\delta+d\delta}\rho u^2 dy\cdot w\right\}dx$$

$\Rightarrow -w\delta dx\cdot\frac{\partial p}{\partial x} - \tau_w\cdot dx\cdot w = w\,dx\,\frac{\partial}{\partial x}\int_0^\delta \rho u^2 dy - Uw\,dx\,\frac{\partial}{\partial x}\int_0^\delta \rho u\,dy \Rightarrow -\delta\frac{dp}{dx} - \tau_w = \frac{\partial}{\partial x}\int_0^\delta \rho u^2 dy - U\frac{\partial}{\partial x}\int_0^\delta \rho u\,dy$

- BERNOULLI $\Rightarrow \frac{dp}{dx} = -\rho U\frac{dU}{dx} \Rightarrow \delta\rho U\frac{dU}{dx} - \tau_w - \frac{\partial}{\partial x}\int_0^\delta \rho u^2 dy - U\frac{\partial}{\partial x}\int_0^\delta \rho u\,dy \Rightarrow LHS = \frac{d}{dx}\int_0^\delta \rho U\,dy$

PRODUCT RULE: $\frac{\partial}{\partial x}\int_0^\delta \rho u U\,dy = \frac{dU}{dx}\cdot\int_0^\delta \rho u\,dy + U\frac{\partial}{\partial x}\int_0^\delta \rho u\,dy$

$\Rightarrow \tau_w = \frac{\partial}{\partial x}\int_0^\delta \rho u U\,dy - U\frac{\partial}{\partial x}\int_0^\delta \rho u\,dy - \frac{\partial}{\partial x}\int_0^\delta \rho u^2 dy + U\frac{\partial}{\partial x}\int_0^\delta \rho u\,dy + \frac{dU}{dx}\int_0^\delta \rho U\,dy - \int_0^\delta \rho u^2 dy - \frac{dU}{dx}\int_0^\infty \rho u\,dy$

$\tau_w = \frac{\partial}{\partial x}\int_0^\infty \rho u(U-u)\,dy + \frac{dU}{dx}\int_0^\delta \rho(U-u)\,dy = \frac{\partial}{\partial x}\left\{U^2\int_0^\delta \rho\frac{u}{U}\left\{1-\frac{u}{U}\right\}dy\right\} + \frac{dU}{dx}\cdot U\int_0^\delta \rho\left(1-\frac{y}{U}\right)dy$

$\Rightarrow \frac{\tau_w}{\rho} = \frac{d}{dx}\left\{U^2\theta\right\} + U\frac{dU}{dx}\cdot\delta^*$ { FINAL EQUATION IN TERMS OF MOMENTUM THICKNESS & DISPLACEMENT THICKNESS !!

- FLAT PLATE: $U = U_0 \Rightarrow \tau_w = \rho U_0^2\frac{d}{dx}\int_0^\delta \frac{u}{U_0}\left\{1-\frac{u}{U_0}\right\}dy$ SCALING $\eta = y/\delta$ $\rightarrow$ $\rho U_0^2\frac{d\delta}{dx}\int_0^1 \frac{u}{U_0}\left\{1-\frac{u}{U_0}\right\}dy$

BCs: @$y=0$, $u=0$, @$y=\delta$, $u=U_0$, @$y=\delta$, $\partial u/\partial y = 0$: ASSUME $u = a+by+cy^2$ $\overset{BCs}{\Rightarrow}$ $u = 2U_0/\delta\cdot y - U_0/\delta^2\cdot y^2$

FROM DEFINITION, $\tau_w = \mu\,du/dy\big|_{y=0} = 2\mu U_0/\delta \Rightarrow 2\mu U_0/\delta = \left(\frac{d\delta}{dx}\right)\cdot\rho U_0^2\int_0^1\left(2\eta - 5\eta^2 + 4\eta^3 - \eta^4\right)d\eta \Rightarrow \delta = \sqrt{\frac{30\mu x}{\rho U_0}}$

EXAMPLE: CONSIDER 2D LAMINAR BOUNDARY LAYER FLOW ALONG A FLAT PLATE. ASSUME THE VELOCITY PROFILE IN THE BOUNDARY LAYER IS SINUSOIDAL, $u/U = \sin(\pi/2\ y/\delta)$. FIND EXPRESSIONS FOR (A) RATE OF GROWTH OF δ AS A FUNCTION OF x. (B) DISPLACEMENT THICKNESS (x) (C) TOTAL FRICTION ON A PLATE OF LENGTH L, WIDTH b.

FOR GIVEN PROFILE, $u=0$ @$y=0$, $u=U$ @$y=\delta$, & $\frac{du}{dy} = \frac{U\pi}{2\delta}\cos\left(\frac{\pi y}{2\delta}\right) = 1$ @$y=\delta$ $\Rightarrow$ CONSISTENT VELOCITY PROFILE

τ_w (BY DEFINITION) $= \mu\cdot\frac{du}{dy}\Big|_{y=0} = \mu\frac{U\pi}{2\delta} = \rho U^2\frac{d}{dx}\int_0^\delta \frac{u}{U}\left\{1-\frac{u}{U}\right\}\cdot dy = \rho U^2\frac{d\delta}{dx}\int_0^1 \sin\left(\frac{\pi}{2}\eta\right) - \sin^2\left(\frac{\pi}{2}\eta\right)d\eta$

$\Rightarrow \frac{\pi\mu U}{2\delta} = \rho U^2\frac{d\delta}{dx}\times\frac{2}{\pi}\left\{1-\pi/4\right\} \Rightarrow \int_\delta \delta\,d\delta = \int_x 11.5\frac{\mu}{\rho U}dx \Rightarrow \delta = \sqrt{\frac{23\mu x}{\rho U}} = \frac{4.8x}{\sqrt{Re_x}}$

DISPLACEMENT THICKNESS (δ^*) $= \int_0^\delta\left(1-\frac{u}{U}\right)dy = \delta\int_0^1\left(1-\sin\frac{\pi}{2}\eta\right)d\eta = \delta\left\{1-2/\pi\right\} \Rightarrow \delta^* = \frac{1.74x}{\sqrt{Re_x}}$

TOTAL FRICTION $= \int_0^L \tau_w\times dx\times b = \int_0^L \rho U^2\frac{d\theta}{dx}dx\cdot b = \rho U^2 b\left\{\theta_L - \theta_0\right\}$

$\theta_L = \int_0^{\delta_L}\frac{u}{U}\left\{1-\frac{u}{U}\right\}dy = \delta_L\int_0^1 \sin\left(\frac{\pi}{2}\eta\right) - \sin^2\left(\frac{\pi}{2}\eta\right)d\eta \Rightarrow F = \frac{0.658\,\rho U^2 bL}{\sqrt{Re_L}}$

- WHEN WE HAVE A TURBULENT BOUNDARY LAYER, THE PROCEDURE REMAINS IDENTICAL. HOWEVER, A COMMON CORRELATION USED IS THE POWER LAW, $u/U = \eta^{1/7}$ (GIVES INFINITE SHEAR AT WALL), SO THIS IS COUPLED WITH A CORRELATION FOR THE WALL SHEAR STRESS.

EXAMPLE: WATER FLOWS AT 2m/s PAST A FLAT PLATE WITH $L = 2m$ IN THE FLOW DIRECTION. THE BOUNDARY LAYER IS TRIPPED, SO IT BECOMES TURBULENT AT THE LEADING EDGE. EVALUATE THE DISTURBANCE THICKNESS δ, DISPLACEMENT THICKNESS δ^*, AND WALL SHEAR STRESS τ_w @ $x=L$. ASSUME A 1/7 POWER TURBULENT VELOCITY PROFILE.